MILITARY SPECIFICATION

BATTERIES, NON-RECHARGEABLE, LITHIUM MANGANESE DIOXIDE

This Specification is approved for use by U.S. Army Laboratory Command. Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers non-rechargeable batteries of the non-reserve type composed of electrochemical cells utilizing a Lithium Manganese Dioxide system (see 6.2).

1.2 Classification.

1.2.1 Type designation. The type designation of Lithium Manganese Dioxide non-rechargeable batteries shall be in the following form. (see 3.1).

BA-	5372	/U
Component (1.2.1.1)	Battery Type number (1.2.1.2)	Installation indicator (1.2.1.3)

1.2.1.1 Component. Non-rechargeable batteries are identified by the two-letter symbol "BA" followed by a hyphen.

1.2.1.2 Battery type number. The batterey type number identifies the basic design of the battery (See 3.1) and consists of a four digit number in the 5001 through 5999 series.

1.2.1.3 Installation Indicator. The installation indicator identifies the equipment the battery is used in; i.e. /PRC-25 or if "Universal", i.e. /U indicates use in various equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SLCET-R, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issures of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder, Tin Alloy; Tin-Lead Alloy; amd Lead Alloy

MILITARY

MIL-W-76	Wire and cable, Hook-up, Electrical, Insulated.
MIL-P-116	Preservation, Methods of.
MIL-W-6858	Welding, Resistance, Spot and Seam.
MIL-F-14072	Finishes for Ground Electronic Equipment.
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base).

STANDARDS

FEDERAL

FED-STD-101	Test Procedures for Packaging Material
FED-STD-595	Colors.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-45662	Calibration System Requirements.

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2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on the date of the solicitation.

AG00000066	Special Packaging Instruction
DOT-E 7052	Transportation Exemption for Lithium Batteries
CFR 49	Code of Federal Regulations Title 49 (Transportation)

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents and other Government documements required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Selection of specification and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 and shall be approved by the procuring activity.

3.3 First article. When specified, samples shall be subjected to first article inspection (see 4.5 and 6.2).

3.4 Materials. Materials listed in Table I, when used, shall be in accordance with the applicable specification (see 4.4 and 6.2).

3.4.1 Metals. All metals which do not enter into the basic electro-chemical reaction of the cell shall resist, or be treated to resist corrosion in accordance with MIL-F-14072.

3.4.1.1 Dissimilar metals. When dissimilar metals which would adversely affect battery performance are used in intimate contact with each oter, protection against electrolysis and corrosion shall be provided in accordance with MIL-F-14072.

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3.4.2 Insulating compounds - flow or shrinking. When tested as specified in 4.8.14 the insulating, impregnating, potting and sealing compounds shall not flow at high temperature, nor crack or draw away from the sides of a container at low temperature. Any compound used shall be non-flammable and non-toxic. Potting shall not inhibit cell vent operations (see 4.7.10).

3.5 Design and construction. Batteries shall be of the design, construction, physical dimensions, weight, and polarity as specified in 3.1.

3.5.1 Intercell connections. Intercell connections between cell blocks and between cell block and terminal shall be so insulated or positioned as to avoid contact with other conducting material. When insulated wire is soldered to terminal lugs, it shall not be bared more than 3/32 inch from the lug, nor shall it extend more than 3/32 inch beyond the lug (see 4.8.17).

3.5.2 Age of cells and batteries. The manufacturer shall certify that the age of the cells and batteries are as follows: (see 6.2)

a. Minimum age of cells, from the time of their fabrication to the time of their presentation for acceptance inspection as batteries, shall be 5 days.

b. Maximum age of cells, from the time of their fabrication to the time of their shipping date, shall be 180 days.

c. Batteries shall be submitted for acceptance inspection not more than 120 days prior to shipping date.

3.5.3 Terminals (insulation resistance).Terminals shall be as specified on the applicable specification sheet (see 3.1) and insulation resistance shall be not less than 5 megohms when tested as specified in 4.8.8.

3.5.3.1 Metal or plastic covers. Metal or plastic covers may be used in construction of cells and batteries. Construction shall be such that it will be impossible for a battery (or any cell of such battery) to be short-circuited by coming in contact with another battery of the same type when placed end to end.

3.5.3.2 Flat contact terminals. Flat contact terminal dimensions shall be as specified on the applicable specification sheet (see 3.1).

If a terminal is flat contact type, but no height dimension is specified, the following test shall apply.

When the battery is placed flat contact down on a flat surface, only the terminal contact shall be in contact with the flat surface. No insulator or cover may protrude out further than the flat contact terminal.

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3.5.4 Safety features. Each cell of the battery shall contain a designed vent. This vent shall function to release over pressure in the cell and thus preclude an explosion. This feature shall operate before the battery temperature reaches 300°F. There shall be a minimum of 100 psi between actuation of the vent and cell explosion (see 4.7.10).

3.5.5 Jackets. (4.7.1) The jackets may consist of either metallic or nonmetallic material. The contents of multicell batteries shall fit snugly enough in the jackets to minimize movement of the cells. Jackets covering one or more cylindrical cells stacked end on end, and having open top and open bottom, shall be so attached to the cells as to prevent them from slipping out when held or shaken vertically.

3.5.5.1 Metallic jackets. The outside of the battery shall have a coating to protect the jacket from corrosion during or at the conclusion of any of the tests specified herein. The contents of the jacket shall be completely insulated from the metal unless otherwise specified. The jacket shall not open at any of its seams after being subjected to the test specified. The test shall be applicable to batteries weighing 5 pounds or more (see 4.7.11.1).

3.5.5.2 Non-metallic jackets. Non-metallic jackets, cardboard or plastic, shall not support combustion nor emit toxic vapors when subjected to flame. The seams of the jacket shall not be open prior to or after being subjected to any test to which the battery is subjected (see 4.7.1).

3.5.5.2.1 Jacket integrity (paperboard). Paperboard jackets shall not fall apart and the seams shall remain intact when tested as specified in 4.7.11.2.

3.5.5.3 Color of Jackets.The color of exposed surfaces of jackets shall match on of the following lusterless greens 34079, 34086, 34087, 34096, 34102, 34127, and 34128 per FED-STD-595 (6.2).

3.5.6 Manufacturing processes. The contractor shall maintain records on all manufacturing processes he employs for the preparation, fabrication and/or refining of cell and battery components. These records shall be available to the Government upon request. The records shall be so maintained that any cell lot and battery fabricated from the cell lot can be identified with respect to material used, batch number, components supplier, and other pertinent process data (see 6.2).

3.5.6.1 Time order production. Battery lots shall be packaged and shipped as produced consecutively (see 6.2).

3.6 Battery voltages.

3.6.1 Battery open-circuit voltage. The open-circuited voltage shall not exceed the maximum voltage specified. (see 3.1 and 4.7.2.1).

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3.6.2 Battery closed-circuit voltage. The closed-circuit voltage shall be not less than the minimum voltage specified (see 3.1 and 4.7.2.2).

3.7 Capacity. When the battery is tested for capacity as specified in 4.7.9 the time required to reach its specified minimum voltage shall be not less than the battery capacity requirement specified (3.1).

3.7.1 Initial voltage delay. When the battery is tested for capacity, the time required at the beginning of discharge for the battery units to reach minimum voltage after the load is applied shall be no more than the time specified (3.1) (see 4.7.9.1.1).

3.8 Electrolyte leakage (cell test). Each cell used in a battery shall have a leakage rate not to exceed 1.0 percent of the total electrolyte in the cell when tested in accordance with 4.7.12.

3.9 Vibration. After the batteries have been tested as specified in 4.7.6 they shall meet the visual and mechanical and battery voltage requirements (see 3.5 and 3.6). There shall be no voltage fluctuations during the test.

3.10 Mechanical shock. After the batteries have been tested as specified in 4.7.5 they shall meet the visual and mechanical and battery voltage requirements (see 3.5 and 3.6). There shall be no voltage fluctuations during the test.

3.11 Drop test. Following the battery drop test, the socket shall remain within the limits specified on the individual specification sheet (see 3.1). The battery shall meet the electrical requirements specified. Following each drop, cells shall not be exposed and battery dimensions shall remain within tolerance (see 4.7.3).

3.12 Altitude. After the batteries have been tested as specified in 4.7.7 they shall meet the visual and mechanical and battery open circuit voltage requirements (see 3.5 and 3.6.1).

3.13 Labeling and marking. All labeling and marking shall be clear and legible throughout all the tests specified herein. Labeling and marking shall be black. Metallic and plastic jackets may have the labeling and marking engraved, or die stamped, in which case it may be the same color as the background.

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3.13.1 Labels. Each battery shall have a label as specified in 3.1. If there is insufficient space to show all required information on one face of the battery, it shall be continued on another face. There shall be no information on the label other than the following:

BATTERY, NON-RECHARGEABLE, LITHIUM MANGANESE DIOXIDE
Type Designation
(Contract Number) --
(Code) -- (See Note and 3.13.1.1)
Serial Number (See 3.13.1.2)
Manufacturer's name
(Trade name may also be-used)
Manufacturer's plant
DO NOT CHARGE, SHORT CIRCUIT, INCINERATE, OR MUTILATE THIS BATTERY
OTHERWISE BATTERY MAY VENT OR RUPTURE RELEASING TOXIC MATERIALS

EXAMPLE:

BATTERY, NON-RECHARGEABLE, LITHIUM MnO_2
BA-5270/U
DAAB05-83-C-1234
0383B

Serial number 1-2-3
James E. Doe Company

JODOCO
Batteryville, NJ

DO NOT CHARGE, SHORT CIRCUIT, INCINERATE OR MUTILATE THIS BATTERY
OTHERWISE BATTERY MAY VENT OR RUPTURE RELEASING TOXIC MATERIALS

Note: the code may be placed on the bottom of single-cell batteries.

3.13.1.1 Code. The Code shown shall indicate the month, year and week of manufacture of the battery by means of a four-digit number followed by a single letter in which the first two digits shall indicate the year. Months earlier than the tenth month shall be a single digit preceded by "0". "A" shall be used for the first week of the month, "B" for the second week of the month, etc. Sunday shall be considered the first day of a week.

EXAMPLES:

A battery manufactured during the second week of March 1983 will bear the code "0383B".

A battery manufactured during the third week of November 1983 will bear the code "1183C".

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3.13.1.2 Serial number. Each battery produced and furnished under a particular contract shall be given a specific serial number. These serial numbers shall be assigned consecutively beginning with one and continuing therefrom for the entire contract lot of batteries. Serialization of batteries shall be accomplished immediately upon completion of fabrication and prior to sampling for inspection. Serial numbers of batteries are not required on shipping containers or packaging.

3.13.2 Special marking. A black band, extending around approximately one third (1/3) of the battery length, shall be provided on each jacket.The term "LITHIUM BATTERY" shall appear in bold-lusterless green lettering (same as jacket color) on the black portion of the jacket on at least one surface of the battery.

3.13.3 Terminal marking. On batteries having socket-type terminals, all markings such as polarity, voltage, and the voltage unit of battery (A, B, C, etc.) shall appear on the face of the battery bearing the socket. On other type terminals, the terminal markings may appear on the top or the side of the battery, or both. Markings shall indicate clearly the terminals to which they refer.

3.14 Cell water content. Water content of the electrolyte solution in a cell shall not exceed 200 parts per million by weight when tested by the Karl Fischer electrolytes titration method. The contractor shall certify that this limit is not exceeded throughout production (see 6.2 and 4.7.18).

3.15 Workmanship. Batteries shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect their life, serviceability, interchangeability, or appearance in accordance with Requirement 9 of MIL-STD-454.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified in herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test adjustment and inspection facilities. Test and measuring equipment and inspection facilities shall be of sufficient accuracy, quality and quantity to permit performance of the required examinations and tests. Unless otherwise specified herein, all examinations and test shall be performed under ambient temperature, humidity, and atmospheric pressure conditions. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.1.1.1 Instrument accuracy

4.1.1.1.1 Voltmeters and ammeters. All voltmeters and ammeters used in testing the batteries shall be accurate within 1 percent of the full scale value. The voltmeter and ammeter ranges shall be such that all readings are taken on the upper half of the scale. The sensitivity of voltmeters shall be not less than 10,000 ohms per volt.

4.1.1.1.2 Resistor tolerance. During all tests involving discharge through a resistance, such resistance shall be accurate within the following percentages:

	Percent
Up to and including 25,000 ohms	+ 0.5
From above 25,000 ohms to and including 1 megohm	+ 1.0
Above 1 megohm	+ 5.0

In determining the resistance used as a test load, the resistance of all continuosly operating voltmeters shall be considered as part of the specified load.

4.1.1.1.3 Power supplies. Power supplies used for discharges specified herein shall be accurate within ± 1 percent.

4.1.1.1.4 Timing. Timing equipment shall be accurate within 0.5 percent.

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4.2 Classification of inspection. The examination and testing of batteries shall be classified as follows:

a. Materials inspection (4.4).

b. First article inspection (see 4.5).

c. Quality conformance inspection of equipment before packaging (see 4.6).

d. Packaging inspection (see 4.8).

4.3 Inspection conditions. Except as otherwise specified herein, all examinations and tests shall be performed at a temperature of 80 ±20°F.

4.4 Materials inspection. Materials inspection shall consist of verifications by certification from the source that the materials used in fabricating the cells and the batteries are in accordance with applicable requirements prior to such fabrication. In the absence of certification from the source, a certificate of analysis or certified inspection data shall be required as proof of conformance to applicable requirements. (see Table I).

TABLE I - Materials inspection.

Material	Requirement Paragraph	Applicable Specification
Metals and Finishes Solder (SN 40) Solder Flux (type RMA)	3.4.1	MIL-F-14072 QQ-S-571 MIL-F-14256

4.5 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Goverment. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.5.1 Sampling plan. The number of batteries constituting an inspection lot shall be in accordance with Table II.

4.5.2 Inspection routine. First article inspection shall consist of all the examinations and tests in accordance with Table II. One sample battery, untested is to remain at the contractor's plant and is to be available as a standard for comparative purposes.

4.5.2.1 First article data. The first article test plan and test report(s) shall be as specified in the contract or purchase order (see 6.2)

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4.5.2.2 Failure. If one or more sample batteries fail to meet any of the first article requirements and tests, the contractor shall immediately make the remedial changes. The contractor at no additional cost to the government shall be required to submit additional first article samples for reinspection. A description of the corrective action taken or to be taken shall be included in the first article inspection reports.

TABLE II - First article inspection.

Group	No. of Batteries	Examination and Test	Required Paragraph	Method of Test Paragraph
I	20 (20)	Visual & mechanical examination	3.5, 3.13	4.7.1
		Battery voltage	3.6	4.7.2
		Drop test	3.11	4.7.3
		Dimension & weight	3.5	4.7.4
		Mechanical Schock	3.10	4.7.5
		Vibration	3.9	4.7.6
		Altitude	3.12	4.7.7
		Insulation resistance terminals	3.5.3	4.7.8
		Capacity	3.7	4.7.9
IA	10	"I" Test @ 70°F	3.7	4.7.9.1.2
IB	5	"L" Test @-20°F	3.7	4.7.9.1.3
IC	5	"H" Test @ 130°F	3.7	4.7.9.1.4
II	25 (25)	Visual-mechanical	3.5, 3.13	4.7.1
		Battery voltage	3.6	4.7.2
		Capacity	3.7	4.7.9
IIA	10	"HT" Test @ 130°F after 4 weeks storage @ 130°F	3.7	4.7.9.1.6
IIB	10	"LT"Test @-20°F after 4 weeks storage @ 130°F	3.7	4.7.9.1.5
IIC	5	"ILR" Test @ 80°F	3.7	4.7.9.1.9
III	15 (15)	Battery Voltage	3.6	4.7.2
IIIA	10	Safety features	3.5.4	4.7.10
IIIB	15 (5)	Jacket integrity (paperboard)	3.5.6.2.1	4.7.11.2
		Visual mechanical	3.5, 3.13	4.7.1
IV	30	Electrolyte Leakage	3.8	4.7.13
V	1	Untested Sample	----	4.5.2

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4.6 Quality conformance inspection of equipment before packaging.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of Groups A, B, and C inspection. Test equipment for government verification inspection shall be made available by the contractor, if required.

4.6.1.1 Inspection Battery Lot. A lot shall be defined as the quantity of batteries of any one type produced at any one place of manufacture on any one contract during one month or less period, submitted at one time to quality conformance inspection.

4.6.1.1.1 Shipment lot. The shipment lot is the quantity of batteries (exclusive of the number of batteries required as samples) of any one type, delivered in one or more shipment lots, under the terms of any one contract.

4.6.1.1.2 Contract lot. The contract lot is the total of all batteries (exclusive of the number of batteries required as samples) of any one type, delivered in one or more shipment lots, under the terms of any one contract.

4.6.1.2 Group A inspection. Each battery on contract or purchase order shall be 100 percent inspected for conformance to the inspections in the order specified in Table III. Discrete lots shall be formed from batteries that pass this inspection.

TABLE III GROUP A INSPECTION.

Examination and Test	Requirement Paragraph	Method of Test Para.
Visual & mechanical	3.5, 3.13 and table VI	4.7.1
Battery Voltage	3.6	4.7.2

4.6.1.3 Group B inspection. Group B inspection shall consist of the tests specified in Table IV in the order shown. Special procedures for small sample inspection of MIL-STD-105 shall apply. Group B inspection shall be performed on sample units from lots which have been subjected to and passed group A inspection.

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TABLE IV. Group B inspection

Examination and test	Requirement paragraph	Method of test Paragraph	Inspection AQL Level
Subgroup I			
Dimensions & Weight	3.5	4.7.1	0.65% S-1
Insulation resistance	3.5.3	4.7.8	0.65% S-1
Safety feature	3.5.4	4.7.10	0.65% S-1
Subgroup II			
Electrolyte leakage	3.8	4.7.12	0.65% S-1
Subgroup III 1/			
Jacket integrity	3.5.5.2.1	4.7.11.2	2.5% S-1
Visual & mechanical	3.5	4.7.1	2.5% S-1

1/ Performed only for paperboard jacketed batteries.

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4.6.1.4 Group C inspection. Group C inspection shall consist of (a)HT capacity test and (b) LT capacity test. Each capacity test shall be preceded by both the mechanical shock and vibration tests described in paragraphs 4.7.5 and 4.7.6 respectively.

4.6.1.4.1 Sampling plan. Samples shall be selected in accordance with General Inspection Level I from Table I of MIL-STD-105. Fifty percent (50%) shall be used for the LT capacity test.

4.6.1.4.2 Group C failures. An AQL of 6.5 shall be used for failures (a), (b), and (c). No failures shall be allowed for (d).

a. Insufficient service.

b. Excessive initial voltage delay.

c. Battery exceeds dimensional tolerances after discharge.

d. Battery vents, burns or ruptures.

4.6.1.5 Group D inspection. Group D inspection shall be performed at the Government inspection facility on sample batteries in accordance with Table V. Shipment of the lot represented by the sample batteries shall not be held up pending the results of group D inspection.

TABLE V - Group D inspection.

Storage and Capacity Test	Requirement Paragraph	Method of Test Paragraph
Group I (Adjustment Purposes see 4.7.1.5.3.1)		
Capacity T	3.7	4.7.9.1.7
Capacity D	3.7	4.7.9.1.8
Capacity ILR	3.7	4.7.9.1.9

4.6.1.5.1 Sampling plan. Random samples for the Group D capacity tests shall be as specified in Table VII.

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TABLE VI - Classification of visual and mechanical examination defects.

Categories	Defects
101	Improper assembly causing parts to be inoperative or unsafe in service.
102	Deformed or damaged parts which are inoperative or malfunction in service.
103	Cell aging requirement not met.
104	Contact surfaces obstructed by insulation material so that electrical use is affected.
105	Torn non-metallic jackets - any tear or rip with dimesion greater than 1/2 inch.
106	Improper jacket closure.
107	Insulating parts or materials missing, damaged, or improperly located so as to affect electrical performance.
108	Location, polarity and marking of terminals not as specified.
109	Labeling and marking wrong, missing or illegible so that utilization is affected.
110	External and internal threads missing, wrong size or so damaged to prevent proper use.
111	Electrolyte leaking caused by missing or defective sealing or closing.
112	Welded or soldered connections improperly made so as to adversely affect battery performance.
113	Improper assembly which could reduce efficiency of operation but not render battery inoperative or unsafe in service.
114	Deformed or damaged parts which do not adversely affect electrical performance.
115	Burns or imperfections which do not interfere with proper use in operation, assembly or disassembly, or cause unsafe condition in service.
116	Improper marking which doesn't hamper utilization or identification of the battery.

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TABLE VII. Sample size and allowable failures for each capacity test of group D.

Inspection Battery Lot Size	Sample size for indicated Group D Capacity Test from Inspection Battery Lots			Allowable Failures 1/		
	T	D	ILR	T	D	ILR
111 to 500 2/	15	15	10	3	3	2
501 to 800	25	25	10	5	5	2
801 to 1300	35	35	10	7	7	2
1301 to 3200	50	50	10	9	9	2
3201 to 8000	75	75	15	13	13	3
8001 to 22,000	110	110	22	18	18	4
22,000 to 110,000	150	150	30	24	24	6
over 110,000	225	225	45	34	34	9

1/ When the number of capacity values falling below the minimum requirements specified (see 3.1) for a given test is equal to or less than the associated allowable failures, the inspection battery lot from which the sample was drawn has met the requirements of that test.

2/ Determination of compliance specified in 4.6.1.5.2 shall not apply to lot sizes of less than 111.

4.6.1.5.2 Compliance. The entire inspection battery lot shall be considered as complying when the T, D, and ILR test results show conmpliance as specified in Table VII.

4.6.1.5.3 Noncompliance.

4.6.1.5.3.1 Group I. If the capacity test results do not show conpliance with the requirements as defined in 4.7.1.5.2, the entire inspection battery lot shall be considered as not complying with requirements of this specification and an adjustment (6.2) can be made if it is determined to be in the Government's best interest by the contracting officer.

4.6.2 Quality conformance inspection of packages. Quality conformance inspection shall be in accordance with MIL-P-116. When required (see 6.2) the rough handling requirements shall be in accordance wth MIL-P-116.

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4.7 Test Methods and examinations.

4.7.1 Visual and mechanical examination. Batteries and cells shall be examined to determine compliance with all applicable requirements and characteristics as specified herein (see 3.5, 3.13, and Table VI.).

4.7.2 Battery Voltage.

4.7.2.1 Open-circuit voltage. A direct current voltmeter of appropriate range and sensitivity shall be used to measure the open-circuit voltage (see 3.6.1).

4.7.2.2 Closed-circuit voltage. A direct current voltmeter of proper range and sensitivity shall be used to measure the closed circuit voltage utilizing the load specified (see 3.6.2).

4.7.3 Drop test. Each battery shall be dropped once for each temperature, from a height of 30±2 inches onto a hard surface consisting of concrete. The side of the battery perpendicular to and nearest the socket shall be parallel to the concrete surface upon release and shall make contact with it. The drop test shall be performed at 160°F and 20°F. The batteries shall be stabilized a minimum of 4 hours at each test temperature and dropped within 10 minutes after removal from the temperature chamber (see 3.11).

4.7.4 Dimensions and weight. Batteries shall be examined by gauging or measuring and by weighing to determine conformance (see 3.5).

4.7.4.1 Dimensions. All dimensions shall include any coating which may be used , and shall remain within the specified tolerances throughout the required tests. Both minimum and maximum dimensions shall be determined. When box gauges are used, batteries, loaded with the following weights shall pass freely through the applicable gauge openings:

a. Batteries weighing less than 5 pounds - loading weight of 5 pounds.

b. Batteries weighing 5 pounds or more - loading weight equal to the weight of the battery.

The dimensions of the box guage shall be the specified maximum outside dimensions of the battery.

4.7.5 Mechanical shock. Each battery shall be secured to the testing machine by means of a rigid mount which shall support all mounting surfaces of the battery. Each battery shall be subjected to a total of three shocks of equal magnitude. The shocks shall be applied in each of three mutually perpendicular directions. Each shock shall be

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applied in a direction normal to a face of the battery. The faces of the battery are identified by their position in relation to the front face (the face which bears the label or marking). For each shock, the battery shall be accelerated in such a manner that during the first 3 milliseconds the minimum average acceleration is 75 gravity units (G). The peak acceleration shall be between 125 to 175G, exact value shall be recorded (see 3.10).

4.7.6 Vibration. Each battery shall be rigidly clamped to the platform of a vibration machine in a manner approximating as closely as practicable the manner in which the batteries are clamped when in use (see 3.1). A simple harmonic motion shall be applied having an amplitude of 0.03 inch (0.06 - inch total maximum excursion). The frequency shall be varied at the rate of 1 hertz per minute between the limits of 10 and 55 cycles per second. The entire range of frequencies and return shall be traversed in 95±5 minutes for each mounting position, (direction of vibration) of the battery. The batteries shall be vibrated in three equal periods in mutually perpendicular directions, one of which shall be perpendicular to the terminal face of the battery. Open circuit voltage shall be observed for 30 seconds during the last quarter of each of the three vibration periods (see 3.9).

4.7.7 Altitude. Batteries shall be placed in an altitude chamber, in which the pressure is maintained at a value corresponding to an altitude of 50,000 feet and the temperature is kept at 75±5°F for a period of six (6) hours (see 3.12). Open circuit voltage only shall be observed after the test.

4.7.8 Insulation resistance (terminals). Insulation resistance test shall be performed, except as otherwise specified (see 3.1). Batteries shall be stored for a period of 48 hours at +70±5°F and a relative humidity of 50±15 percent. After storage and while at these conditions, the insulation resistance shall be measured by applying a direct-current potential of 500±20 volts between any two terminals not electrically connected and between all ungrounded terminals and the container of the battery. The insulation resistance of batteries having a non-metallic container shall be measuired by the use of a 1 inch-square copper plate making physical contact with the container. The plate shall be placed with the broad surface against any area of any surface of the jacket other than that on which the battery terminals are located (see 3.5.3).

4.7.9 Capacity (see 3.7).

4.7.9.1 Capacity Tests. Sample batteries selected for capacity tests specified in the individual specification sheet (3.1) shall be stored and discharged in air as applicable in accordance with 4.7.9.1.1 through 4.7.9.2.1. All batteries shall be discharged to 1.0 volts, except those subjected to the ILR test. The time required to reach the specified minimum voltage shall be used to determine the battery capacity.

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4.7.9.1.1 Initial voltage delay. At the start of the capacity discharge test, batteries shall be monitored with an oscillographic device or equivalent to determine the time in milliseconds required for battery units to reach minimum voltage after the specified loads are applied as specified in the individual specifications sheets. No loads shall be applied at any time during storage or prior to the discharge test following such storage (see 3.7.1).

4.7.9.1.2 Capacity test I. Discharge at +70±5°F without previous storage.

4.7.9.1.3 Capacity test L. Discharge at -20±3°F after storage at-20±3°F for a minimum of sixteen (16) hours.

4.7.9.1.4 Capacity test H. Discharge at +130±3°F after storage at +130±3°F for a minimum of sixteen (16) hours.

4.7.9.1.5 Capacity test LT. Discharge at -20±3°F after four (4) weeks storage at 130°F followed by a minimum of sixteen (16) hours at -20±3°F. No load shall be applied during the storage at 130°F and -20°F or prior to discharge test.

4.7.9.1.6 Capacity test HT. Discharge at +130±3°F after four (4) weeks storage at 130F° and a minimum of sixteen (16) hours at +130±3°F.

4.7.9.1.7 Capacity test T. Discharge at +70±5°F after thirteen (13) weeks storage at +130°F.

4.7.9.1.8 Capacity test D. Discharge at +70±5°F after fifty-two (52) weeks storage at +70°F.

4.7.9.1.9 Capacity test ILR. Discharge at +80±20°F to the specified minimum voltage without previous storage.

4.7.9.2 Storage conditions. The storage conditions specified in Table VIII shall prevail during storage periods specified. Normal conditions shall be maintained. No loads shall be applied at any time during storage or prior to the discharge test following storage. Deviations from normal conditions are permitted provided that (1) the extreme conditions specified in Table VIII do not exist for more than ten percent (cumulative of the specified storage periods) and (2) that at no time are the extreme conditions exceeded. Batteries must be oriented in storage to have at least 50 percent of the cell seals at the bottom of the batteries (see 3.1).

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Table VIII Storage test conditions

Kinds of storage	Normal conditions temperature	Extreme Conditions temperature
T, HT, and LT storage	+130 + 3°F - 4°F	108°F thru 126°F and 133°F thru 135°F
D Storage	+70 ± 5°F	60°F thru 65°F and 75°F thru 80°F
Temperature Cycling	+160 + 3°F - 7°F	140°F thru 153°F and 163°F thru 165°F

4.7.9.2.1 Discharge. Following stabilization the batteries shall be discharged at the ambient discharge conditions as specified. There shall be at least a two (2) inch separation between all batteries that are being discharged. All batteries subjected to capacity testings shall be discharged as specified in individual specification sheets. A battery shall be considered to have failed a discharge test when any one of the following conditions occurs:

a. The battery voltage or the voltage of any one unit falls below the specified minimum voltage prior to exceeding the capacity or fails to reach minimum voltage in the required time.

b. The battery dimensions exceed the maximum specified (see 3.1).

c. Any evidence of cell venting, rupture or fire is observed.

d. Insufficient service.

4.7.10 Safety feature. Fifty percent of batteries set aside for safety feature test shall be tested according to paragraph 4.7.10.1. The remaining batteries shall be tested according to paragraph 4.7.10.2.

4.7.10.1 Short circuit test. Each battery shall be subjected to a direct short, i.e., less than .02 ohms, for a period of time not less than 24 hours or until the battery temperature returns to ambient. At the end of this test, the battery must be checked that no cell vented. Batteries containing cells which exploded, burned, or vented are failures. Distortion of labeling or dimensions is permitted.

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4.7.10.2 Vent test. All batteries shall be placed at +167°F (75) for two hours. At the end of the two hours the battery must be checked to see that no cell has vented. If any cell has vented, the battery has failed this test. If no cell has vented, the ambient temperature shall be raised to a maximum of 300°F. Batteries shall be maintained at an ambient of +300°F for two hours.

All cells must have vented at the designated vent after the two hours at 300°F. If the cell has a crimp seal construction, (ie: rubber, plastic), then venting is also allowed at the crimp seal. If the cell is hermetically sealed, then venting is only permitted at the designed vent.

Batteries containing cells which exploded, burned or did not vent are failures (see 3.5.4). Distortion of labeling or dimensions is permitted.

4.7.11 Jackets.

4.7.11.1 Metallic jackets. Metallic jacketed batteries weighing five pounds or more shall be loaded by applying weights totaling 100 pounds evenly distributed over the top of the battery and shall remain so loaded at least one minute. The condition of the jacket shall be observed.

4.7.11.2 Jacket integrity (paperboard). Batteries shall be immersed to within 1/4 inch of the top of the jacket in water maintained at a temperatue of +113±5°F for a period of 48 hours. The condition of the jacket and its seams shall be observed (see 3.5.5.2.1). This test is only to be conducted on paperboard jackets.

4.7.12 Electrolyte leakage test. At no time shall there be potting substances or cell jackets applied to the cells designed for this test. The cells will be numbered and weighed to the nearest milligram. The weight will be recorded and the amount of electolyte in each cell known. Fifty percent of these cells will be stored for four weeks at 130°F. The remaining cells will be temperatured cycled. They will be cycled for 12 hours at +160°F (71°C) and 10 hours at -60°F (-51°C) and 1 hour in between to change temperatures. This 24 hour cycle will be repeated 7 times for a total time of 7 days.

At the end of each test, the cells will be removed and allowed to cool at room temperature for at least two hours. Each cell shall then be weighed to the nearest milligram. The weight loss must be recorded. The weight loss shall not exceed 1.0 percent of the total electrolyte weight. The electrolyte salt weight shall not be used in this computation.

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4.7.12.1 Sample Size. The number of cells examined per inspection battery lot shall equal the same number of batteries subjected to the "T" capacity test per inspection battery lot.

4.7.12.2 Failure Rate. The failure rate shall be the same as for the "T" capacity test in Table VII.

4.7.13 Flow or shrinking (insulating compounds) (see 3.4.2). Compounds shall be placed in a container, approximately 3 inches wide by 6 inches long by 3/4 inch high, to within 1/4 inch of the top. The temperature of the compound within the container shall be raised to +200±5°F and the container shall be held in an inverted position for 24 hours. Then the temperature of the compound shall be lowered to -40±5°F.

4.7.14 Intercell connections. Intercell connections shall be visually examined for conformance to 3.5.1.

4.7.15 Cell water content. Cell water content shall be certified for first article inspection in the first article test report and for each inspecion battery lot in the quality conformance test report. (see 3.14 and 6.2).

4.8 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.

b. Quality Conformance Inspection of Packaging.

4.8.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.8.2 Quality Conformance Inspection of Packaging.

4.8.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.8.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.8.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table IX, "PACKING INSPECTION PROVISIONS." Lot formation shall consist all packs made of the materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality Level of 4.0 percent defective.

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TABLE IX .PACKING INSPECTION PROVISIONS

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specifications	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specifications	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with AG00000066.

5.2 Shipments. Department of Transportation DOT-E-7052 or applicable CFR 49 in effect at the time of shipments shall apply. Batteries shall not require metal drums as the outside shipping container for air shipment.

6. NOTES

6.1 Intended use. The primary batteries included are of the non-reserve type composed of electrochemical cells utilizing a lithium manganese dioxide system. The lithium cell batteries are capable of storage and use under wide temperature ranges.

6.1.1 Indirect shipments. The packaging, packing and marking specified in Section 5 apply only to direct purchases by or direct shipment to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Applicable specification sheet.

c. Complete type designation (see 1.2.1).

d. Requirement for first article test plan and test report (see 4.5.2.1).

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e. Requirement for maintaining manufacturing processes (see 3.5.6).

f. Requirement for certification of materials and processes (see 3.4).

h. Requirement for certification of age of cells (see 3.5.2).

i. Requirement for contractor to state color of jackets (see 3.5.6.3).

j. Adjustment procedures for Group D inspection for Capacity (see 4.6.1.5.3).

k. When first article rough handling tests are not required.

l. Requirement for certification of cell water content (see 3.14).

m. Level A or B preservation and packing (see section 5).

n. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.3 Definitions

6.3.1 Non-flammable and non-toxic materials. Non-flammable and non-toxic materials are those materials which will not support combustion, produce smoke, or be capable of emitting toxic fumes when subjected to the environmental conditions specified for the battery.

6.4 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.5 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.6 International standardization agreement. Provision of this specification are the subject of international standardization agreements.When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standarization Offices so that appropriate action may be taken respecting the international agreement concerned.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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